

EX-6

DELUGE SCRIPT – BASICS

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1. Aim

To study the basics of Deluge scripting in Zoho Creator and understand how Deluge can be used with variables, data types, operators, expressions, conditional statements, Lists, and Maps to implement application logic.

2. Introduction

Deluge (Data Enriched Language for the Universal Grid Environment) is the scripting language used in Zoho applications to automate tasks and implement application logic. It provides simple programming constructs for storing values in variables, performing calculations, working with data types, using operators and expressions, and controlling program flow. Deluge scripts can be used to process data, perform calculations, validate information, and automate actions in Zoho Creator. The Deluge Script Basics activity provides practice through challenges covering Variables, Data Types, Operators, Expressions, Comments, Conditionals, List, and Map.

3. Objectives

- To understand the basic syntax and structure of Deluge scripts.
- To learn how to declare and use variables in Deluge.
- To understand data types and operators used in Deluge expressions.
- To perform calculations and create expressions using variables.
- To learn the use of conditional statements for decision-making.
- To understand how Lists and Maps can be used to organize data.
- To develop basic problem-solving and automation skills using Deluge.

4. Advantages

- Easy-to-understand syntax makes Deluge suitable for beginners.
- Helps automate repetitive tasks in Zoho applications.
- Supports calculations, expressions, conditions, and data processing.
- Can be used to implement customized application logic.
- Works closely with Zoho Creator forms, reports, and application components.
- Reduces manual work by automating application processes.

5. Disadvantages

- Deluge is mainly designed for the Zoho ecosystem and is not a general-purpose language.
- Advanced programming requirements may be limited compared with full programming languages.

- Complex scripts can become difficult to maintain if they are not properly organized.
- Users need knowledge of Zoho application components to build effective automation.
- Debugging complex scripts may require careful testing and understanding of Deluge functions.

6. Result

The Deluge Script Basics activity was completed successfully. The exercise provided an understanding of variables, data types, operators, expressions, conditionals, Lists, and Maps in Deluge. It also demonstrated how simple Deluge scripts can be used to perform calculations and implement basic logic for Zoho applications.

Level up your Zoho Deluge scripting skills — 60 challenges across Variables, Data Types, Operators, Expressions, Comments, Conditionals, List & Map — live scoring, and a certificate at the finish line.

LATIKA SHIVANI N Score: 14.4 /100 Streak: 0 Q 16 / 60

Data Types EASY

Declare a variable `department` holding a Text (String) data type value `"AI & DS"`.

Your Deluge Script copy / paste / right-click disabled

```
department="IT&cse"
```

✗ **Not quite right.** Compare your script with the model answer below and keep going.

Model answer: `department = "AI & DS";`

[Next Question →](#)

LATIKA SHIVANI N

Score: 14.4 /100

Streak: 1

Q 15 / 60

Expressions

MEDIUM

Write an expression that checks whether `num` is even, storing the boolean result in `isEven`, using the modulus operator.

Your Deluge Script

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```
iseven=num%2==0;
```

✓ **Correct!** +1.6 points. Great job on **Expressions**.

Model answer: `isEven = (num % 2 == 0);`

Next Question

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Score: 12.8 /100

Streak: 6

Q 13 / 60

Operators

MEDIUM

Write a logical expression using AND (`&&`) that checks `age` is greater than 18 and `hasID` is true, storing result in `canVote`.

Your Deluge Script

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```
canvote=age>18 && hasID;
```

✓ **Correct!** +1.6 points. Great job on **Operators**.

Model answer: `canVote = (age > 18 && hasID == true);`

Conditionals, List & Map — live scoring, and a certificate at the finish line.

LATIKA SHIVANI N

Score: **12.8** /100

Streak: **0**

Q **14** / 60

HARD

Sort the list `marksList` in ascending order and store the result in `sortedMarks`.

Your Deluge Script

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```
sortedmark=markslist;  
sortedmark.sort(true);
```

✗ **Not quite right.** Compare your script with the model answer below and keep going.

Model answer: `sortedMarks = marksList.sort(true);`

Next Question →

Deluge Script - Basics

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Score: **10.1** /100

Streak: **4**

Q **11** / 60

Operators

MEDIUM

Use the NOT operator (`!`) to negate the boolean variable `isActive` and store the result in `isInactive`.

Your Deluge Script

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```
isinactive =!isActive;
```

✓ **Correct!** +1.6 points. Great job on **Operators**.

Model answer: `isInactive = !isActive;`

Next Question →

Level up your Zoho Deluge scripting skills — 60 challenges across Variables, Data Types, Operators, Expressions, Comments, Conditionals, List & Map — live scoring, and a certificate at the finish line.

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Score: 11.2 /100

Streak: 5

Q 12 / 60

Variables

EASY

Declare a variable named `collegeName` and assign it the text value `"REC"` .

Your Deluge Script

copy / paste / right-click disabled

```
collegename="rec";
```

✓ **Correct!** +1.1 points. Great job on **Variables**.

Model answer: `collegeName = "REC";`

Level up your Zoho Deluge scripting skills — 60 challenges across Variables, Data Types, Operators, Expressions, Comments, Conditionals, List & Map — live scoring, and a certificate at the finish line.

LATIKA SHIVANI N

Score: 6.9 /100

Streak: 2

Q 9 / 60

Data Types

EASY

Declare a variable `rollNo` holding a Number (integer) data type value `101` .

Your Deluge Script

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```
rollno=101;
```

✓ **Correct!** +1.1 points. Great job on **Data Types**.

Model answer: `rollNo = 101;`

Next Question →

Level up your Zoho Deluge scripting skills — 60 challenges across Variables, Data Types, Operators, Expressions, Comments, Conditionals, List & Map — live scoring, and a certificate at the finish line.

LATIKA SHIVANI N

Score: 5.9 /100

Streak: 1

Q 8 / 60

Comments

MEDIUM

Write a multi-line comment (using `/*` and `*/`) that says: `This function calculates the result based on marks .`

Your Deluge Script

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```
/*
this function calculates the result based on marks.
*/
```

✓ Correct! +1.6 points. Great job on Comments.

Model answer: `/* This function calculates the result based on marks */`

Next Question →

Level up your Zoho Deluge scripting skills — 60 challenges across Variables, Data Types, Operators, Expressions, Comments, Conditionals, List & Map — live scoring, and a certificate at the finish line.

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Score: 4.3 /100

Streak: 0

Q 7 / 60

Comments

EASY

Write a single-line comment stating `declaring student variables` directly above a variable declaration line.

Your Deluge Script

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```
student name="harishni";
```

✗ Not quite right. Compare your script with the model answer below and keep going.

Model answer: `// declaring student variables studentName = "Arun";`

Next Question →

Level up your Zoho Deluge scripting skills — 60 challenges across Variables, Data Types, Operators, Expressions, Comments, Conditionals, List & Map — live scoring, and a certificate at the finish line.

LATIKA SHIVANI N

Score: 4.3 /100

Streak: 0

Q 6 / 60

Map

MEDIUM

Find the number of key-value pairs in the map `student` and store it in `fieldCount`.

Your Deluge Script

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```
fielcount=student.size();
```

✗ **Not quite right.** Compare your script with the model answer below and keep going.

```
Model answer: fieldCount = student.size();
```

Next Question →

Conditionals, List & Map — live scoring, and a certificate at the finish line.

LATIKA SHIVANI N

Score: 4.3 /100

Streak: 1

Q 5 / 60

Operators

MEDIUM

Use the string concatenation operator (`+`) to join `"Roll No: "` with the variable `rollNo`, storing it in `label`.

Your Deluge Script

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```
label="roll no:"+rollno;
```

✓ **Correct!** +1.6 points. Great job on **Operators**.

```
Model answer: label = "Roll No: " + rollNo;
```

LATIKA SHIVANI N

Score: 2.7 / 100

Streak: 0

Q 4 / 60

List

HARD

Remove the element "Maths" from the list `subjects`.

Your Deluge Script

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```
subject.remove("maths");
```

✗ Not quite right. Compare your script with the model answer below and keep going.

```
Model answer: subjects.removeElement("Maths");
```

Next Question →